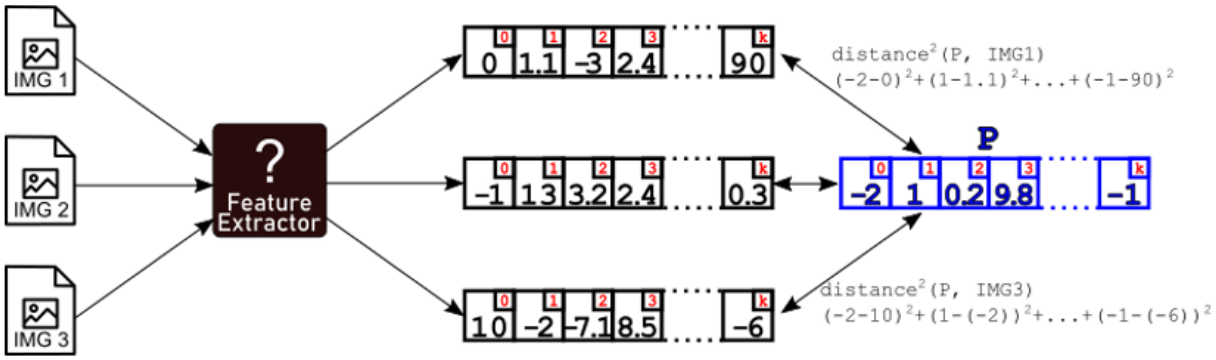


Figure 10.1 Feature extraction on an image dataset. Each image is translated into a feature vector (though what's represented as a "black box" feature extractor, as we are not interested in the algorithm that creates this vectors). Then, if we have to search an entry P, we compare P's feature vector to each of the images' vectors, computing their mutual distance based on some metric (in figure, Euclidean distance – notice that, when looking for the minimum of these Euclidean distances, we can sometimes compute the squared distances, avoiding to apply a square root operation for each entry).

Contrary to the feature extraction, the search algorithms haven't changed much in the last ten years, and the data structures that we introduce in this chapter, invented between the late 90s and the early 2000s, are still cutting-edge choices for efficient search in the vector space.

10.1.2 Overcoming k-d trees Flaws

Back in chapter 9, we also mentioned a couple of possible structural solutions to cope with the problems mentioned in the previous section: